Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Physical vector spaces for quantum mechanics

Files.

- File: `briefs/QuantumVectorSpaces/QuantumVectorSpaces.lean`
- Library: `QuantumVectorSpaces`
- Imports: `SchwartzNorms` (the L^2 seminorm family on $\mathcal{S}(\mathbb{R})$) and two new `ForMathlib` files: `ForMathlib/Analysis/InnerProductSpace/Defs.lean` (the seminorm $\sqrt{\mathrm{re}\langle x, x\rangle}$ of a `PreInnerProductSpace.Core`, and continuity of the inner product against it), `ForMathlib/Analysis/CStarAlgebra/Matrix.lean` (`Matrix.toEuclideanLin` is multiplicative)

Setting. V is a complex vector space with an inner product given as a `PreInnerProductSpace.Core`. Unlike Mathlib's `InnerProductSpace`, this puts no norm or topology on V. The defining observables are a family X_i ($i \in \iota$) in `Module.End`, and `algebra` is the unital algebra they generate (`Algebra.adjoin`). The seminorms $p_A(v) = \sqrt{\langle Av, Av\rangle}$ form `seminormFamily`, and `topology` is the topology they generate. A topology is `IsAdmissible` if it makes V a topological vector space

and makes the X_i and the inner product (jointly) continuous. For Example 3, $Xf = xf$ and $Pf = -i\hbar f'$ act on Mathlib's $\mathcal{S}(\mathbb{R}, \mathbb{C})$ (`posOp`, `momOp`), with the L^2 inner product `l2Core`. For Example 4, $S_i = (\hbar/2)\sigma_i$ acts on $\mathbb{C}^2$ as `EuclideanSpace` (`spin`).

Results.

Brief	Lean
Corollary 2, coarsest topology	`isGreatest_topology` (from `isAdmissible_topology` and `IsAdmissible.continuous_algebra`)
Corollary 2, locally convex	`locallyConvexSpace`
finite dimension (Example 4)	`topology_eq_moduleTopology`, `topology_qubit`
Example 3, $[X, P] = i\hbar$	`posOp_mul_momOp_sub`
Example 3, X, P, I symmetric	`isSymmetric_obs`
Example 3, Remark: the topology is that of $\mathcal{S}$	`topology_schwartz`, `withSeminorms_schwartz`, `isAdmissible_schwartz` (for $\hbar \neq 0$)
Example 4, $[S_i, S_j] = i\hbar\,\epsilon_{ijk} S_k$	`spin_comm`
Example 4, S_i symmetric	`isSymmetric_spin`
Theorem 5, [B]COBS	`IsAdmissible.continuous_expectation`

Differences from the brief.

- "Coarsest" is `IsGreatest` in Mathlib's order on topologies, where coarser is larger.
- The inner product is only assumed positive semidefinite in Corollary 2; definiteness is used only in finite dimension, to make the topology Hausdorff.
- Example 3 is formalized for $\mathcal{S}(\mathbb{R})$ only, not $\mathcal{S}(\mathbb{R}^n)$. The proof that the topologies agree has two halves. (a) Each seminorm $\|x^a \partial^b f\|_2$ equals $|\hbar|^{-b} p_{X^a P^b}(f)$. (b) Every element of the algebra is a combination of the normally ordered monomials $X^a P^b$ (`algebra_le_monoSpan`, from $PX^{n+1} = X^{n+1}P - (n+1)i\hbar X^n$), so every p_A is bounded by finitely many $\|x^a \partial^b f\|_2$. With `withSeminorms_l1_l2` from the previous brief, this gives the Schwartz topology.
- In finite dimension the topology of Corollary 2 is the unique Hausdorff vector-space topology, whatever the defining observables (`topology_eq_moduleTopology`). This is how Example 4 is checked.
- Not formalized: completeness and second countability in Definition 1 (Mathlib has no completeness of $\mathcal{S}$ at this pin); the "iff $n = m$" part of Example 3, which needs a notion of isomorphism of quantum vector spaces; [B]WDQ and [B]STDIST in Theorem 5 and the remarks of section 5, which are informal.

Problems found in the brief. No errors. Two steps are left implicit.

- The Remark after Example 3 says the L^2 seminorms and the sup seminorms dominate each other. But the seminorms of Corollary 2 are $\|Af\|$ for every A in the algebra, including non-normally-ordered products such as PX. Reducing them to $\|x^a \partial^b f\|$ needs the commutation relation (normal ordering), which the brief does not mention.
- Example 3 needs $\hbar \neq 0$: for $\hbar = 0$ we get $P = 0$, and the topology is just that of L^2.